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Using mlxconfig

Using mlxconfig with PCI Device in Bus Device Function (BDF) Format

In order to access device in BDF format via configuration cycles, use "pciconf-" as prefix to the device.

Example:

```

1  # mlxconfig -d 0000:86:00.0 q
2
3  Device #1:
4  -----
5  Device type:    ConnectX5
6  Name:          MCX556A-ECA_Ax
7  Description:    ConnectX-5 VPI adapter card; EDR IB (100Gb/s) and 100GbE; du
8  Device:        0000:86:00.0
9
10 Configurations:
11     MEMIC_BAR_SIZE                0
12     MEMIC_SIZE_LIMIT              _256KB(1)
13     HOST_CHAINING_MODE            DISABLED(0)
14     HOST_CHAINING_CACHE_DISABLE   False(0)
15     HOST_CHAINING_DESCRIPTOR      Array[0..7]
16     HOST_CHAINING_TOTAL_BUFFER_SIZE Array[0..7]
17     FLEX_PARSER_PROFILE_ENABLE    0
18     FLEX_IPV4_OVER_VXLAN_PORT     0
19     ROCE_NEXT_PROTOCOL            254
20     ESWITCH_HAIRPIN_DESCRIPTOR    Array[0..7]
21     ESWITCH_HAIRPIN_TOT_BUFFER_SIZE Array[0..7]
22     PF_BAR2_SIZE                  0
23     NON_PREFETCHABLE_PF_BAR       False(0)
24     VF_VPD_ENABLE                 False(0)
25     STRICT_VF_MSIX_NUM            False(0)
26     VF_NODNIC_ENABLE              False(0)
27     NUM_OF_VFS                    0
28     PARTIAL_WRITE_CACHE_MODE       DEVICE_DEFAULT(0)
29     PF_BAR2_ENABLE                False(0)
30     SRIOV_EN                      False(0)
31     PF_LOG_BAR_SIZE               5
32     .....

```

Using mlxconfig to Set IB/ETH Parameters

In order to set IB/ETH parameters through mlxconfig, use the following command line:

```
1 # mlxconfig -d <device> set [LINK_TYPE_P1=<link_type>] [LINK_TYPE_P2=<link_t
```

Example: Configuring both ports as InfiniBand:

```
1 # mlxconfig -d /dev/mst/mt4119_pciconf0 set LINK_TYPE_P1=1 LINK_TYPE_P2=1
2
3 Device #1:
4 -----
5 Device type:    ConnectX5
6 Name:          MCX556A-ECA_Ax
7 Description:    ConnectX-5 VPI adapter card; EDR IB (100Gb/s) and 100GbE; du
8 Device:        /dev/mst/mt4119_pciconf0
9
10 Configurations:
11 LINK_TYPE_P1    Next Boot    New
12 LINK_TYPE_P2    ETH(2)       IB(1)
13
14 Apply new Configuration? (y/n) [n] : y
15 Applying... Done!
16 -I- Please reboot machine to load new configurations.
```

Using mlxconfig to Set SR-IOV Parameters

In order to set SR-IOV parameters through mlxconfig, use the following command line:

```
1 # mlxconfig -d <device> set [SRIOV_EN=<0|1>] [NUM_OF_VFS=<NUM>]
```

Example: Turning on SR-IOV and enabling 8 Virtual Functions per Physical Function:

```
1 # mlxconfig -d /dev/mst/mt4115_pciconf0 set SRIOV_EN=1 NUM_OF_VFS=8
2
3 Device #1:
4 -----
5 Device type:    ConnectX4
6 PCI device:    /dev/mst/mt4115_pciconf0
7 Configurations:    Next Boot    New
8 SRIOV_EN          0            1
9 NUM_OF_VFS        0            8
10
11 Apply new Configuration? ? (y/n) [n] : y Applying... Done!
12 -I- Please reboot machine to load new configurations.
```

Using mlxconfig to Set Preboot Settings

For a full description of the preboot configurable parameters refer to [Supported Configurations and their Parameters](#) under "Preboot Settings".

Example: Enable boot option ROM on port 1, set boot retries to 3 and set the boot protocol to PXE (on ConnectX-3 Pro cards only).

```
1 # mlxconfig -d /dev/mst/mt4103_pciconf0 set BOOT_OPTION_ROM_EN_P1=1 BOOT_RET
2
3 Device #1:
4 -----
5 Device type:    ConnectX3Pro
6 PCI device:    /dev/mst/mt4103_pciconf0
7 Configurations:      Next Boot      New
8   BOOT_OPTION_ROM_EN_P1   False(0)    True(1)
9   BOOT_RETRY_CNT_P1       0            3
10  LEGACY_BOOT_PROTOCOL_P1  2            1
11
12 Apply new Configuration? ? (y/n) [n] : y
13 Applying... Done!
14 -I- Please reboot machine to load new configurations.
```

Example: Configure VLAN ID to 3 on port 2

```
1 # mlxconfig -d /dev/mst/mt4119_pciconf0 set BOOT_VLAN=3
2
3 Device #1:
4 -----
5 Device type:    ConnectX5
6 Name:          MCX556A-ECA_Ax
7 Description:    ConnectX-5 VPI adapter card; EDR IB (100Gb/s) and 100GbE; du
8 Device:        /dev/mst/mt4119_pciconf0
9
10 Configurations:      Next Boot      New
11   BOOT_VLAN          1            3
12
13 Apply new Configuration? (y/n) [n] : y
14 Applying... Done!
15 -I- Please reboot machine to load new configurations.
```

Using mlxconfig to Split a Port in a Remotely Managed Switch

The break-out cable is a unique NVIDIA capability, where a single physical quad-lane QSFP port is divided into 2 dual-lane ports. It maximizes the flexibility of the end user to use the NVIDIA switch with a combination of dual-lane and quad-lane interfaces according to the specific requirements of its network. All system ports may be split into 2-lane ports. Splitting a port changes the notation of that port from x/y to x/y/z with “x/y” indicating the previous notation of the port prior to the split and “z” indicating the number of the resulting sub-physical port (1,2). Each sub-physical port is then handled as an individual port. For example, splitting port 5 into 2 lanes gives the following new ports: 1/5/1 & 1/5/2.



To enable the port split, the following actions are required:

Step 1. Set the Split Mode in a Remotely Managed Switch.

```
1 # mlxconfig -d <device>set SPLIT_MODE=1
```

Example:

```
1 # mlxconfig -d /dev/mst/mt54000_pciconf0 set SPLIT_MODE=1
2
3 Device #1:
4 -----
5 Device type:    Quantum
6 Name:          N/A
7 Description:    N/A
8 Device:        /dev/mst/mt54000_pciconf0
9 Configurations: Next Boot          New
10 SPLIT_MODE     NO_SPLIT_SUPPORT(0) SPLIT_2X(1)
```

To create a query for the Split Mode parameter using mlxconfig:

```
1 # mlxconfig -d <device> q SPLIT_MODE
```

Example:

```
1 # mlxconfig -d /dev/mst/mt54000_pciconf0 q SPLIT_MODE
2
3 Device #1:
4 -----
5 Device type:    Quantum
6 Name:          N/A
7 Description:    N/A
8 Device:        /dev/mst/mt54000_pciconf0
9 Configurations: Next Boot
10 SPLIT_MODE     SPLIT_2X(1)
```

Step 2. Split a Port in a Remotely Managed Switch.

- To split a specific port for one or more ports of (1-64) using mlxconfig:

```
1 # mlxconfig -d <device> set SPLIT_PORT[<port_num>/<port_range>]=1
```

Please note that although the command, is “set SPLIT_PORT[33..64]”, it splits a specific port for one or more ports of the higher ports (33-40).

Please note that the first port is set as 1, e.g., [1], [1..64].

- How to turn on the split port for the **first port** only:

```
1 mlxconfig -d /dev/mst/SW_MT54000_EVB-SX-1_L00_lid-0x0001 set SPLIT_PORT
```

- How to turn on the split port for the **first 32 ports** (range of ports):

```
1 mlxconfig -d /dev/mst/SW_MT54000_EVB-SX-1_L00_lid-0x0001 set SPLIT_PORT
```

- How to turn on the split port for the **last 8 ports**:

```
1 mlxconfig -d /dev/mst/SW_MT54000_EVB-SX-1_L00_lid-0x0001 set SPLIT_PORT
```

- How to turn off the split port for **port 40**:

```
1 mlxconfig -d /dev/mst/SW_MT54000_EVB-SX-1_L00_lid-0x0001 set SPLIT_PORT
```

- How to query the split port for the **first 32 ports**:

```
1 mlxconfig -d /dev/mst/SW_MT54000_EVB-SX-1_L00_lid-0x0001 query SPLIT_PC
```

Step 3. Reboot the switch.

To disable the port split, the following actions are required:

Step 1. Disable the Split Ports in a Remotely Managed Switch:

- To unsplit a specific port for one or more ports (1-32) using mlxconfig:

```
1 # mlxconfig -d <device> set SPLIT_PORT[<port_num>/<port_range>]=0
```

Step 2. Disable the Split Mode in a Remotely Managed Switch.

```
1 # mlxconfig -d <device> set SPLIT_MODE=0
```

mlxconfig Raw Configuration Files

mlxconfig allows applying raw configuration file for a pre-set configuration. Raw configuration files are intended for advanced users. This document does not cover the generation of such files.

Set the raw configuration file:

```
1 # mlxconfig -f ./tlv_file.conf -d /dev/mst/mt4115_pciconf1 set_raw
2
3 Raw TLV #1 Info:
4 Length: 0xc
5 Version: 0
6 OverrideEn: 1
7 Type: 0x00000080
8 Data: 0xa0000000 0xa0020010 0x00000000
9
10 Raw TLV #2 Info:
11 Length: 0x4
12 Version: 0
13 OverrideEn: 1
14 Type: 0x01020012
15 Data: 0x0000000b
16
17 Raw TLV #3 Info:
18 Length: 0x8
19 Version: 0
20 OverrideEn: 1
21 Type: 0x00000190
22 Data: 0x00000010 0x00007d00
23
24 Raw TLV #4 Info:
25 Length: 0x4
26 Version: 0
27 OverrideEn: 1
28 Type: 0x00000082
29 Data: 0x0000000c
30 Operation intended for advanced users.
31
32 Are you sure you want to apply raw TLV file? ? (y/n) [n] : y
33 Applying... Done!
34 -I- Please reboot machine to load new configurations.
```

Never apply files from an unreliable source.



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